
Survival Prediction under Distribution Shift

QRT Data Challenge — Overall Survival Prediction for patients
diagnosed with Myeloid Leukemia

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TL;DR

First online data-science competition.

Public leaderboard

0.7685

Rank 11 / 743

Private leaderboard

0.7124

Rank 189

The gap between those two numbers is the most interesting thing in this work.

The problem

Predict overall survival in MDS / AML patients from clinical + genetic features.

Inputs

- Clinical: BM blast %, WBC, hemoglobin, platelets
- Cytogenetics: ISCN-format karyotype string
- Molecular: per-mutation records (gene, effect, VAF, protein change)

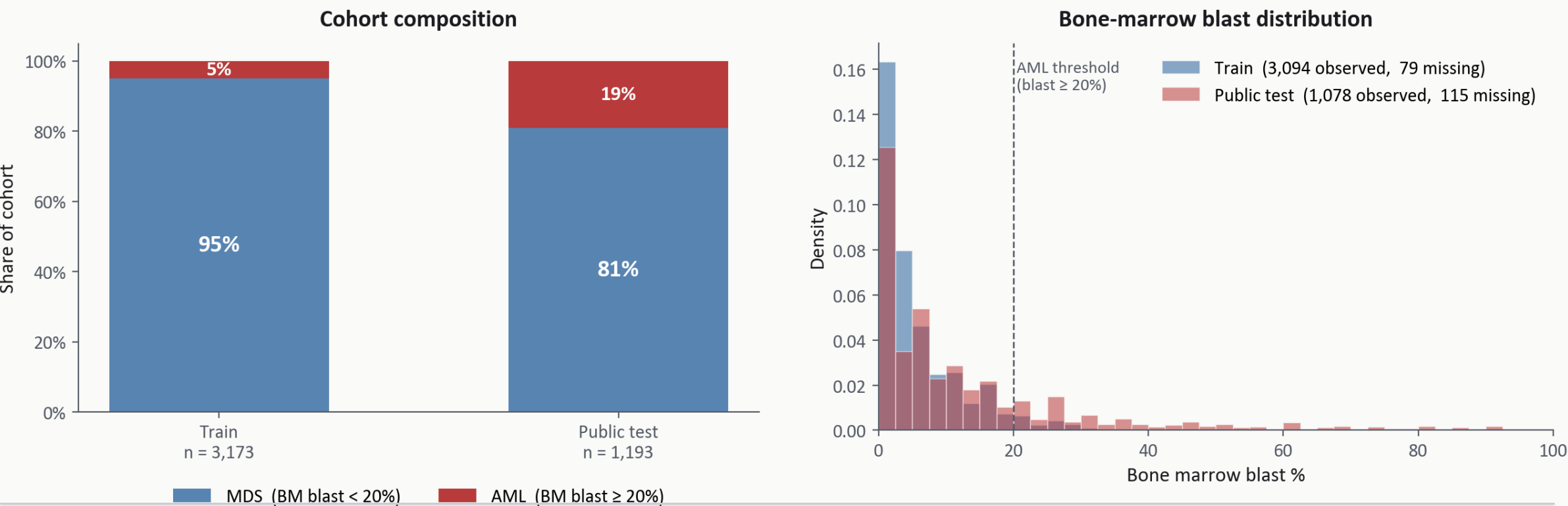
Target & metric

- (OS_YEARS, OS_STATUS) — time-to-event survival outcome
- IPCW-C: censoring-aware concordance index
- Rewards correct pairwise risk orderings, not point predictions

Constraint: 2 leaderboard submissions per day. This will matter later.

Distribution shift: cohort composition and features

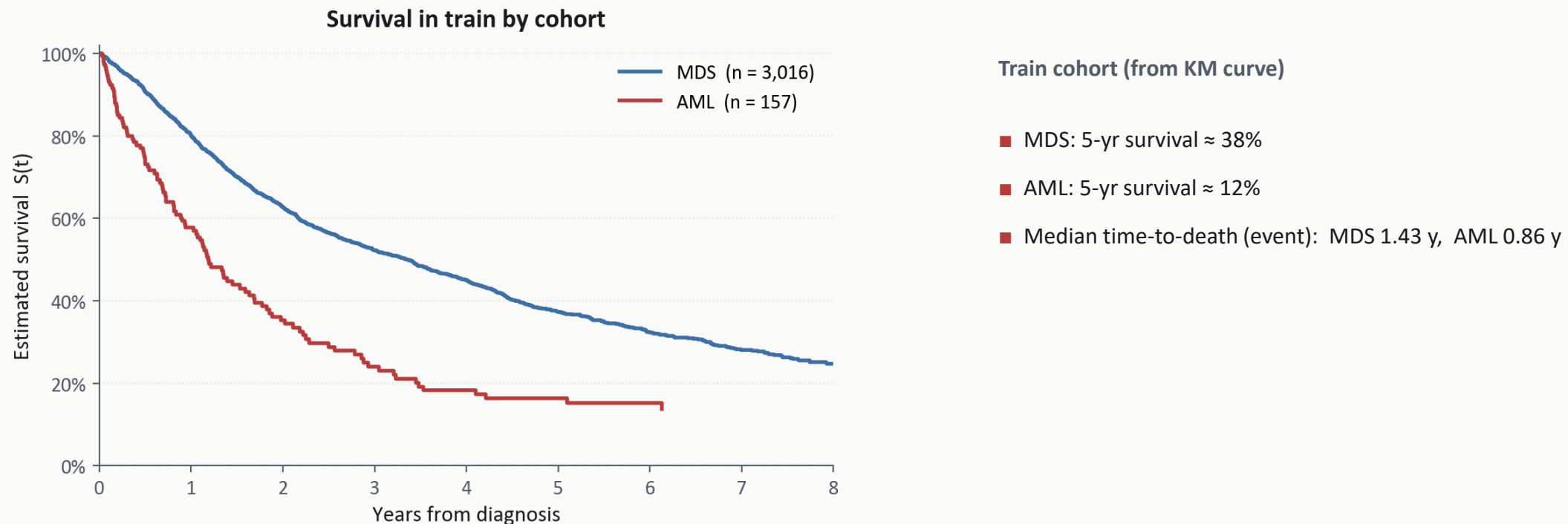
Train and test were drawn from different mixtures of the same disease spectrum.



Public test has ≈4× the AML share of train (5% → 19%); median blast doubles (3% → 6%).

...and so do the survival outcomes those features predict

AML patients die faster — so a test set enriched for AML is implicitly enriched for early-death labels.



IPCW-C rewards correct ordering among high-risk patients — the very ones the test set is enriched for.

Pipeline: two complementary survival models

BASELINE (v12)

Wide features, single RSF

- ~1800 features: gene one-hots × VAF, cyto one-hots, EFFECT, PROTEIN_CHANGE
- Dense IPSS-M score (Bernard 2022 NEJM Evidence)
- Random Survival Forest, 5000 trees

Public LB: 0.7637

SPECIALIST (v22)

23 biologically-robust features

- Clinical + curated cyto + curated genes + engineered scores
- Trained on QRT + 583 Tazi patients (propensity-selected)
- Single RSF (500 trees)

Public LB: 0.7645 alone

FINAL = $0.30 \cdot \text{rank}(\text{baseline}) + 0.70 \cdot \text{rank}(\text{specialist}) \rightarrow \text{Public LB: 0.7685}$

rank-blend (not raw): two RSF architectures live on different scales, $\sim[218,2039]$ vs $\sim[127,1204]$

Baseline detail: feature engineering

Combining five sources of signal into ~1800 columns.

- Clinical numerics — log1p transform compresses right-skewed counts (WBC, PLT) so the tree splits aren't dominated by extreme values; MICE imputation iteratively regresses each variable on the others (RandomForest as regressor) to preserve cross-feature correlations that median-fill would destroy.
- Per-gene one-hot weighted by max VAF — captures both presence (was the gene mutated?) and allelic burden (how clonal is the mutation?).
- Two cytogenetic parsers — a broad regex extracts every +/-, translocation and inversion (high recall, surfaces rare lesions); a curated parser produces clean IPSS-R-aligned flags (high precision).
- EFFECT and PROTEIN_CHANGE — mutation type (missense, ITD, PTD, ...) and residue position parsed from HGVS strings like p.R882H.
- IPSS-M score (Bernard 2022) — the full Molecular International Prognostic Scoring System as a single dense feature. Carries ~42% of tree-ensemble importance.
- Two interaction flags — NPM1+/FLT3- (favorable AML), and TP53+ × complex karyotype (high-risk MDS/AML).

Specialist: cross-cohort augmentation via propensity selection

Pull external patients matched to the QRT high-blast test region.

How the propensity classifier works

- Label QRT test patients with BM blast $\geq 20\%$ as 1; label all Tazi 2022 NEJM Evidence cohort patients as 0
- Train LightGBM 5-fold to discriminate, on raw inputs only (no engineered scores)
- Sort Tazi by $P(\text{label}=1)$; keep the top 583 — most similar to underserved AML region

Why K = 583

Sized to match the public test's AML share. Adding 583 high-blast Tazi patients to the 3,173 QRT train cohort lifts the augmented train AML share from $\approx 5\%$ to $\approx 19\%$ — the same AML share as the public test set.

Why raw inputs only

Engineered features (IPSS-M, TP53 multihit) inside the propensity model would correlate selection criteria with the specialist's own scoring — a feedback loop that narrows the augmented distribution. Worth +0.0011 public LB.

Rank-blending: scale-invariant complementarity

The two models output incompatible scales:

Baseline range: ~[218, 2039]

Specialist range: ~[127, 1204]

→ Raw averaging lets the baseline dominate by scale alone.

Convert to ranks, then blend.

final = 0.30 · rank(baseline) + 0.70 · rank(specialist)

Also matches IPCW-C's geometry — the metric scores pairwise orderings, not magnitudes.

Ablation: the augmentation is the load-bearing element

Each piece earning its keep on the public LB.

Variant	Spec training	α	Public LB	Δ vs v12
v12 baseline alone	—	—	0.7637	—
Specialist alone	QRT + Tazi-583	—	0.7645	+0.0008
Spec + v12, NO augmentation	QRT only	0.50	0.7584	-0.0053
Spec + v12, with augmentation	QRT + Tazi-583	0.50	0.7674	+0.0037
Final (v22, Option C)	QRT + Tazi-583	0.30	0.7685	+0.0048

Apples-to-apples at $\alpha=0.50$: Tazi augmentation alone moves the blend +0.0090 (0.7584 $\rightarrow$ 0.7674).

Without augmentation, the specialist drags the blend BELOW v12 alone (-0.0053).



Public leaderboard: rank 11 / 743

Rank	Score	Team
1	0.7739	ocercle
2	0.7732	qrtmieuxquecitadel
3	0.7732	Astra
...	...	...
11	0.7685	chawkinas (me)

Top of the public leaderboard at 0.7739 — within 0.0054 of my score.



Private leaderboard: a 5-point shake-up across the field

	Public	Private	Drop
Top of LB (rank 1)	0.7739	0.7231	-0.0508
chawkinas (me)	0.7685	0.7124	-0.0561

The entire field shook up — top public 0.7739 → top private 0.7231 (−0.0508).

My drop is 0.0053 worse than the winner's — within ~0.005 of the rest of the top of the public LB.

For reference: organisers' benchmark scored 0.6411 on private (rank 678) — beat it by +0.0713.

Technical lessons

Three findings that transfer beyond this project.

1. Components that fail individually can win in combination.

Specialist alone -0.0040 LB · Tazi augmentation alone -0.0033 · combined $\rightarrow +0.0048$ over v12.

Don't drop a building block just because it loses standalone — multi-component systems with complementary errors can beat the sum of their parts.

2. Feature-schema design enables cross-cohort augmentation.

Same 583 Tazi patients: into baseline's 1,800 sparse features $\rightarrow -0.0033$ LB; into specialist's 23 shared features $\rightarrow +0.0090$ swing.

More data isn't enough — the feature vocabulary has to transfer across cohorts. Plan for transfer at the feature stage, not the model stage.

3. Beware feedback loops in propensity-driven selection.

v21 propensity used engineered scores (the specialist's own signals); v22 used raw inputs only; $+0.0011$ LB from breaking the loop.

If a selection procedure shares features with the model it serves, the model is implicitly choosing its own training data — a subtle selection bias.

What I learned

First DS competition. The methodology I'd defend; the iteration policy I'd change.

- Late-stage optimization against a public test set is itself a form of overfitting.
- When CV and LB disagree, trust CV — especially once per-iteration LB deltas drop below noise band (~ 0.003).
- Pick a final submission once deltas plateau; don't nudge it for marginal LB gains.
- Treat the public LB as a sanity check, not a continuous optimization target.
- A 0.001 improvement on a 1193-patient C-index is within sampling noise. The improvement may not be real.

Thank you.

Code, methodology writeup, post-mortem:

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